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On the slice genus of generalized algebraic knots

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ABSTRACT

We give examples of a linear combination of algebraic knots and their mirrors that are algebraically slice, but whose topological and smooth four-genus is two. Our examples generalize an example of non-slice algebraically slice linear combination of iterated torus knots obtained by Hedden, Kirk and Livingston. Our main tool is a genus bound from Casson–Gordon theory and a cabling formula that allows us to compute effectively these invariants.

Keywords: Algebraic knots; knot concordance; Casson-Gordon invariants.

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1. Introduction

A knot $K \subset S^3$ is *algebraic* if it arises as a link of an isolated singularity of a complex curve. A *generalized algebraic knot* (*GA-knot* for short) is a knot that can be written as a connected sum of algebraic knots and their reverse mirror images.

Rudolph asked [28] whether algebraic knots are linearly independent in the smooth concordance group. By [25, Corollary 8.4], Rudolph's question can be reformulated by asking whether every smoothly slice GA-knot is ribbon. For the sake of brevity, we will refer to the conjecture asserting the affirmative answer to Rudolph's question as *Rudolph's conjecture*.

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Probably the first result toward Rudolph's conjecture is due to Litherland [20]. By studying the jumps of the Levine–Tristram signature function, he proved that torus knots are linearly independent in the smooth concordance group. Livingston and Melvin [23] proved that there are GA-knots that are algebraically slice but not smoothly slice, hence the Levine–Tristram signature is not sufficient to answer Rudolph's question. Later, Hedden, Kirk and Livingston [15] used Casson–Gordon invariants to show that there is an infinite family of algebraically-slice GA-knots, which are not smoothly slice. Their approach was extended by Conway, Kim and the second author [5].

Recall that in dimension four, there is a big difference between smooth and topological (locally flat) concordance of knots and links. Using [26], it can be verified that Litherland's result holds also in the topological concordance group. Similarly, it's been understood for many years that the invariance of Casson–Gordon signatures with respect to topological concordance can be deduced from the results of Freedman-Quinn [12] and Gilmer [13]. Consequently, the independence results discussed in the previous paragraph hold in the topological concordance group as well.

The examples of algebraically slice non-slice GA-knots studied in [15, Sec. 8] have topological 4-genus equal to one. However, the main result of [5] gives infinitely generated subgroups of the topological concordance group consisting of algebraically-slice GA-knots. Therefore, it is an interesting question whether there are examples of algebraically slice GA-knots with arbitrary large topological 4-genus. We propose the following conjecture.

Conjecture 1. *There are algebraically slice GA-knots with arbitrary large topological* 4*-genus.*

The purpose of this paper is to provide some examples supporting Conjecture 1.

Before stating our main theorem, let us introduce some notation. For the rest of the paper, for positive coprime integers p, q, we let $T(p, q)$ denote the (p, q)-torus knot. If also r, s are positive coprime integers, we write $T(p, q; r, s)$ for the (r, s)-cable of $T(p, q)$. Note that every positive torus knot is an algebraic knot. Recall that if $s > pqr$, the knot $T(p, q; r, s)$ is algebraic as well; see [9]. For a knot K, $g_4(K)$ denotes the smooth 4-genus of K and $g_4^{\mathrm{top}}(K)$ denotes the topological 4-genus of K. Finally, let $-K$ denote the reverse mirror image of K.

Theorem 1.1. *Consider the following GA-knot*:

$$K = T(2,17;2,83) \# -T(2,11;2,83) \# T(2,83) \# -T(2,13;2,83)$$
$$\# T(2,11;2,103) \# -T(2,103) \# T(2,13;2,103) \# -T(2,17;2,103).$$

K is algebraically slice and $g_4(K) = g_4^{\mathrm{top}}(K) = 2$.

There are many more examples of algebraically slice GA-knots with $g_4 = g_4^{\mathrm{top}} = 2$. Among them, the example from Theorem 1.1 is the smallest one we

could find. Few other examples are listed below.

$$\begin{aligned}
K_1 &= T(2,23;2,107) \# -T(2,17;2,107) \# T(2,107) \# -T(2,19;2,107) \\
&\quad \# T(2,17;2,131) \# -T(2,131) \# T(2,19;2,131) \# -T(2,23;2,131) \\
K_2 &= T(2,29;2,139) \# -T(2,19;2,139) \# T(2,139) \# -T(2,23;2,139) \\
&\quad \# T(2,19;2,163) \# -T(2,163) \# T(2,23;2,163) \# -T(2,29;2,163) \\
K_3 &= T(2,37;2,163) \# -T(2,29;2,163) \# T(2,163) \# -T(2,31;2,163) \\
&\quad \# T(2,29;2,181) \# -T(2,181) \# T(2,31;2,181) \# -T(2,37;2,181)
\end{aligned}$$

To keep the paper concise we focus on the example given in Theorem 1.1.

Our main tool for establishing the necessary 4-genus bounds is the Casson–Gordon σ-invariant [3, 4]. In particular, we use the results of Gilmer [14], see also [10]. We note that the following tools are insufficient to give any nontrivial 4-genus bounds.

- By the definition, Levine–Tristram signatures fail to provide any 4-genus bounds for an algebraically slice knot.
- For an algebraically-slice GA-knot K we have $\tau(K) = s(K) = 0$. Similarly, all the higher s-invariants coming from $sl(N)$ homologies [19, 22], for $N \geq 2$, are zero. This is proved in [15, Proposition 8.2].
- The Υ invariant of Ozsváth, Stipsicz and Szabó [27] vanishes for K. The argument is as in [5, Sec. 1.2] and involves the result of Tange [31].

The proof of Theorem 1.1 consists of two steps: establishing an upper and a lower bound. To obtain the upper bound, we construct an explicit genus-two surface in B^4 which bounds the knot from Theorem 1.1. This is done in Lemma 4.1. On the other hand, the lower bound is established with the aid of Gilmer's bound [14, Theorem 1] and hence requires much more work. In order to apply Gilmer's lower bound, we had to resort to computer calculations to verify the required assumptions. The calculations were done using a SageMath [29] script, available at [24]. The output of the Sage script is attached to the arXiv version of the paper.

The complexity of calculations required to verify assumptions of Gilmer's lower bound increases significantly if one takes more complicated examples of GA-knots. However, it is possible to make some optimizations by considering only the primary summands of the linking form, see Remark 4.2 for a more detailed discussion. In particular, for the knot from Theorem 1.1, we were able to reduce the number of cases to check from roughly $6 \cdot 10^{12}$ to $1.6 \cdot 10^6$. We made attempts to find examples of algebraically slice GA-knots with topological 4-genus equal to three, however, the calculations were too complicated even including the aforementioned optimizations.

The paper is organized as follows. Section 2 serves as a quick recollection of the most relevant properties of Casson–Gordon σ-invariants and the 4-genus bound of

Gilmer. In Sec. 3 we discuss our implementation of calculations of Casson–Gordon invariants. In Sec. 4, we give the proof of Theorem 1.1.

2. Casson–Gordon Invariants

2.1. *Linking forms*

In this section we review some basic properties of linking forms. For a more detailed study of the algebra of linking forms refer to [16, 33].

A *linking form* is a pair (A, λ) consisting of a finite abelian group A and a nonsingular symmetric bilinear pairing

$$\lambda\colon A \times A \to \mathbb{Q}/\mathbb{Z}.$$

Here, nonsingularity means that the adjoint map

$$\lambda^D\colon A \to \hom(A, \mathbb{Q}/\mathbb{Z}), \quad \lambda^D\colon x \mapsto (y \mapsto \lambda(x, y))$$

is an isomorphism of abelian groups.

Example 2.1. Fix a prime $q \in \mathbb{Z}$. For a non-zero integer a we will denote by (a/q) the symmetric form given by the formula

$$(a/q)\colon \mathbb{Z}_q \times \mathbb{Z}_q \to \mathbb{Q}/\mathbb{Z}, \quad (x, y) \mapsto \frac{a}{q} \cdot x \cdot y.$$

Observe that (a/q) is nonsingular (and hence a linking form) only if $\gcd(a, q) = 1$.

Example 2.2. Let K be a knot in S^3 and let $\Sigma(K)$ denote the double branched cover of K. By [1, Proposition 8.20], $\Sigma(K)$ is a rational homology sphere, hence there is a linking form

$$\lambda_K\colon H_1(\Sigma(K)) \times H_1(\Sigma(K)) \to \mathbb{Q}/\mathbb{Z}.$$

According to [2, Sec. 2.2], the adjoint of λ_K is equal to the composition

$$\lambda_K^D\colon H_1(\Sigma(K)) \xrightarrow{PD} H^2(\Sigma(K)) \xrightarrow{B} H^1(\Sigma(K); \mathbb{Q}/\mathbb{Z}) \xrightarrow{\text{ev}} \hom(H_1(\Sigma(K)), \mathbb{Q}/\mathbb{Z}).$$

In the above diagram, PD denotes the inverse of the Poincaré duality isomorphism given by the cap product with the fundamental class of $\Sigma(K)$, B denotes the inverse of the Bockstein map associated to the short exact sequence $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$, and ev is the evaluation map coming from the Universal Coefficient Theorem. Since $\Sigma(K)$ is a rational homology sphere, the maps B and ev are isomorphisms. Consequently, λ_K is indeed nonsingular.

For two linking forms (A_1, λ_1) and (A_2, λ_2), we define their (orthogonal) sum $(A_1 \oplus A_2, \lambda_1 \oplus \lambda_2)$ by the formula

$$(\lambda_1 \oplus \lambda_2)((x_1, x_2), (y_1, y_2)) = \lambda_1(x_1, y_1) + \lambda_2(x_2, y_2).$$

For $n > 0$, we will denote by $n\lambda_1$ the n-fold direct sum of the linking form λ_1. For a subgroup $M \subset A$ define its *orthogonal complement*

$$M^{\perp} = \{x \in A : \forall_{y \in M}\ \lambda(x, y) = 0\}.$$

We say that $M \subset A$ is *isotropic* if $M \subset M^{\perp}$ and a is *metabolizer* if $M^{\perp} = M$. Similarly, we say that $x \in M$ is *isotropic*, if $\lambda(x, x) = 0$. The linking form (A, λ) is *metabolic* if it admits a metabolizer.

For a symmetric integral $n \times n$ matrix H with $\det(H) \neq 0$, consider the symmetric bilinear form

$$\lambda_H : (\mathbb{Z}^n/H\mathbb{Z}^n) \times (\mathbb{Z}^n/H\mathbb{Z}^n) \to \mathbb{Q}/\mathbb{Z}, \quad \lambda_H(x + H\mathbb{Z}^n, y + H\mathbb{Z}^n) = x^T \cdot H^{-1} \cdot y.$$

We say that the linking form (A, λ) is *represented* by the symmetric integral matrix H, if (A, λ) is isomorphic to $(\mathbb{Z}^n/H\mathbb{Z}^n, \lambda_H)$.

If (A, λ) is a linking form, the underlying abelian group A admits a primary decomposition

$$A = \bigoplus_p A_p,$$

where the above direct sum is taken over all primes and A_p denotes the p-primary part of A. According to [33, Sec. 5], the primary decomposition of A gives a primary decomposition of λ

$$(A, \lambda) = \bigoplus_p (A_p, \lambda_p), \tag{1}$$

where λ_p denotes the restriction of λ to A_p.

Lemma 2.3. *Suppose that (A, λ) is a linking form admitting a metabolizer M, then M admits a primary decomposition*

$$M = \bigoplus_p M_p,$$

where $M_p = M \cap A_p$ is a metabolizer for the linking form (A_p, λ_p), for any prime p.

Proof. Choose a metabolizer $M \subset A$ for λ and let $M_p = M \cap A_p$, for any prime p. Observe that $M_p^{\perp_{\lambda_p}} = M^{\perp_\lambda} \cap A_p$, where $M^{\perp_\lambda}$ and $M^{\perp_{\lambda_p}}$ denote orthogonal complements with respect to λ and λ_p, respectively. Indeed, the inclusion $M^{\perp_\lambda} \cap A_p \subset M_p^{\perp_{\lambda_p}}$ is obvious. On the other hand, if $x \in A_p$, then by the primary decomposition (1), we have $\lambda(x, y) = 0$, for any $y \in A_q$, for any prime $q \neq p$. Hence $M_p^{\perp_{\lambda_p}} \subset M^{\perp_\lambda} \cap A_p$. Therefore,

$$M_p^{\perp_{\lambda_p}} = M^{\perp_\lambda} \cap A_p = M \cap A_p = M_p,$$

which proves that M_p is a metabolizer for λ_p, as desired. □

2.2. *Characters*

For an integer $m > 1$, let $C_m \subset S^1$ denote the multiplicative subgroup generated by primitive roots of unity of order m. Similarly, by C_∞ we denote the subgroup of S^1 consisting of roots of unity, i.e. C_∞ is generated by the set $\bigcup_{m\geq 1} C_m$.

If A is a finite abelian group, a *character* on A is a homomorphism

$$\chi\colon A \to C_\infty.$$

For a fixed $m \geq 1$, we say that the character χ is an *order m character*, if $\chi(A) = C_m$. Observe that the set $\mathrm{Char}(A)$ of characters on a finite abelian group A can be identified with the set $\hom(A, \mathbb{Q}/\mathbb{Z})$. Indeed, any $\phi \in \hom(A, \mathbb{Q}/\mathbb{Z})$ corresponds to the character given by the formula

$$\chi(\phi)(y) = \exp(2\pi i\phi(y)). \tag{2}$$

The following lemma states a relation between linking forms and character groups.

Lemma 2.4. *Let (A, λ) be a linking form, then λ determines a natural identification of abelian groups*

$$T_\lambda\colon A \cong \mathrm{Char}(A),$$

where, for $x \in A$, $T_\lambda(x) = \chi_x$, where $\chi_x(y) = \exp(2\pi i\lambda(x, y))$.

Proof. As we observed above, there is an identification $\hom(A, \mathbb{Q}/\mathbb{Z}) \cong \mathrm{Char}(A)$ given by formula (2). Furthermore, by nonsingularity of λ, we obtain a natural identification $\lambda^D\colon A \cong \hom(A, \mathbb{Q}/\mathbb{Z})$, hence the lemma follows. We can define T_λ to be the composition of these isomorphisms. □

2.3. *Background on Casson–Gordon invariants*

Fix a positive integer $m > 0$. Recall from the previous section that C_m denotes the subgroup of S^1 generated by primitive roots of unity of order m. A topological *n-manifold over C_m* (or a topological n-dimensional C_m-manifold) is a pair (M, χ), where M is a compact, oriented topological n-manifold and $\chi\colon H_1(M) \to C_m$ is a homomorphism.

We say that a C_m-manifold (M, χ) of dimension n *bounds over C_m* if there exists a topological $(n + 1)$-manifold (W, ψ) over C_m such that

$$\partial(W, \psi) = (M, \chi).$$

In particular, we require that M is the oriented boundary of W and $\chi = \psi \circ \iota$, where $\iota\colon M \hookrightarrow W$ is the inclusion map. Furthermore, we say that two closed topological n-manifolds (M_1, χ_1), (M_2, χ_2) over C_m are *bordant over C_m* if the C_m-manifold

$(M_1 \sqcup -M_2, \chi_1 \oplus \chi_2)$ bounds over C_m. For a positive integer r, we will use the abbreviated notation

$$r \cdot (M, \chi) = \left(\bigsqcup_r M, \bigoplus_r \chi \right).$$

The following lemma is crucial for defining Casson–Gordon invariants of knots and links in the topological category. The lemma is known to the experts in the field, however, in the literature, the lemma is stated for smooth manifolds only, see [7, Proposition 2.12] or [4, p. 183 below the example]. Hence, to fill this gap, we provide a sketch of the proof which works in the topological category.

Lemma 2.5. *Fix an integer $m > 0$. Let (M, χ) be a closed 3-dimensional C_m-manifold, then there exists an integer $r > 0$ and a 4-dimensional C_m-manifold (W, ψ) such that*

$$r \cdot (M, \chi) = \partial(W, \psi),$$

i.e. $r \cdot (M, \chi)$ bounds over C_m.

Sketch of the proof. We only sketch the argument. More details can be found in [17, Chap. 10–11 of Annex C], [18], see also [30] or [8].

Recall that $\Omega_3^{\mathrm{STOP}}(C_m)$ denotes the bordism group of closed oriented topological 3-dimensional C_m-manifolds. In other words, $\Omega_3^{\mathrm{STOP}}(C_m)$ is the set of equivalence classes of homeomorphisms classes of closed oriented 3-dimensional C_m-manifolds under the relation of bordism over C_m. Disjoint sum equips $\Omega_3^{\mathrm{STOP}}(C_m)$ with the structure of an abelian group (the empty manifold is the neutral element). In particular, a C_m-manifold $r \cdot (M, \chi)$, where $r > 0$, represents the trivial element in $\Omega_3^{\mathrm{STOP}}(C_m)$ if it bounds over C_m. Consequently, it is sufficient to prove that $\Omega_3^{\mathrm{STOP}}(C_m)$ is a finite abelian group.

In the smooth category, oriented bordism groups $\Omega_n^{\mathrm{SO}}(C_m)$ of smooth manifolds are identified with homotopy groups of the spectrum $\mathrm{BC}_m \wedge \mathrm{MSO}$, where BC_m is the classifying space of the group C_m and MSO denotes the oriented Thom spectrum, see [8, Chap. 8.7] The identification is done via the Pontriagin–Thom construction. Once the identification is made, bordism groups can be computed with the aid of the Atiyah–Hirzebruch spectral sequence, see [8, Sec. 9.2] for sample calculations.

In the topological category, the topological transversality, see [17], [12, Theorem 9.5A] or [11, Chap. 10], can be used to make the Pontriagin–Thom construction work. In particular, we can identify the topological oriented bordism groups, see [18, Lecture 3],

$$\Omega_n^{\mathrm{STOP}}(C_m) \cong \pi_n^S(\mathrm{BC}_m \wedge \mathrm{MSTOP}),$$

MSTOP denotes the relevant Thom spectrum, see [8, Chap. 8.7]. Stable homotopy groups $\pi_n^S(\mathrm{BC}_m \wedge \mathrm{MSTOP})$ can be calculated with the aid of the Atiyah–Hirzebruch spectral sequence, refer to [8, Sec. 9.2]. In the end, we obtain an isomorphism $\Omega_3^{\mathrm{STOP}}(C_m) \cong C_m$, which concludes the proof. □

Let (W, ψ) be a 4-dimensional topological C_m-manifold with boundary. The cyclic group C_m acts by multiplication on the complex plane, hence we can consider twisted homology groups $H_*(W, \mathbb{C}^\psi)$ and $H_*(\partial W, \mathbb{C}^\chi)$, where χ denotes the restriction of ψ to ∂W. Recall that the ψ-twisted Poincaré duality can be used to construct the ψ-twisted intersection form of W

$$Q_\psi \colon H_2(W, \mathbb{C}^\psi) \times H_2(W, \mathbb{C}^\psi) \to \mathbb{C},$$

see [32, Appendix D.6]. The ψ-twisted intersection form is hermitian with respect to the complex conjugation. We will denote by $\operatorname{sign}^\psi(W)$ the signature of Q_ψ. Similarly, we will denote by $\operatorname{sign}(W)$ the untwisted signature of W, i.e. $\operatorname{sign}(W) := \operatorname{sign}^{\psi_0}(W)$, where ψ_0 is the trivial homomorphism.

Definition 2.6 ([3, 4]). Fix a knot K and a character $\chi \colon H_1(\Sigma(K)) \to C_m$. The *Casson–Gordon σ-invariant* and the *Casson–Gordon nullity* of K are defined by the formulas

$$\sigma(K, \chi) = \frac{1}{r}(\operatorname{sign}^\psi(W) - \operatorname{sign}(W)),$$
$$\eta(K, \chi) = \dim_{\mathbb{C}} H_1(\Sigma(K), \mathbb{C}^\chi),$$

where (W, ψ) is a 4-dimensional C_m-manifold bounding $r \cdot (\Sigma(K), \chi)$, for some positive integer r.

Remark 2.7. Observe that Lemma 2.5 guarantees the existence of the required four-dimensional C_m-manifold in Definition 2.6. Furthermore, we can use the argument from [3] (see the discussion just before Theorem 1 in the paper) coupled with computations of topological bordism groups to conclude, that $\sigma(K, \chi)$ is well-defined, i.e. it does not depend on the choice of a 4-manifold.

Basic properties of the σ-invariant and nullity are summarized in the following proposition.

Proposition 2.8. *Let K be a knot in S^3.*

(1) *If χ is trivial, then*

$$\sigma(K, \chi) = \eta(K, \chi) = 0.$$

(2) *If K_1 and K_2 are knots with characters χ_1 and χ_2, then*

$$\sigma(K_1 \# K_2, \chi_1 \oplus \chi_2) = \sigma(K_1, \chi_1) + \sigma(K_2, \chi_2),$$
$$\eta(K_1 \# K_2, \chi_1 \oplus \chi_2) = \begin{cases} \eta(K_1, \chi_1) + \eta(K_2, \chi_2) + 1 & \chi_1, \chi_2 \text{ nontrivial}, \\ \eta(K_1, \chi_1) + \eta(K_2, \chi_2) & \text{otherwise}. \end{cases}$$

Proof. Let us first prove Item 1. Suppose that χ is the trivial character on $H_1(\Sigma(K))$. Let W be a compact, connected, oriented smooth 4-manifold bounding $\Sigma(K)$. Since the 3-dimensional oriented bordism group $\Omega_3^{\mathrm{SO}}(pt)$ is trivial, such W always exists.

Let $\psi_0\colon H_1(W) \to C_m$ be the trivial character. Since χ is trivial, $\partial(W, \psi_0) = (\Sigma(K), \chi)$ in $\Omega_3^{\mathrm{STOP}}(C_m)$. By definition,

$$\sigma(K, \chi) = \mathrm{sign}^{\psi_0}(W) - \mathrm{sign}(W) = \mathrm{sign}(W) - \mathrm{sign}(W) = 0,$$

since the triviality of ψ_0 implies that $\mathrm{sign}^{\psi_0}(W) = \mathrm{sign}(W)$.

Similarly,

$$\eta(K, \chi) = \dim_{\mathbb{C}} H_1(\Sigma(K), \mathbb{C}^{\chi}) = \dim_{\mathbb{C}}(\Sigma(K); \mathbb{C}) = 0.$$

The middle equality above follows from the fact that χ is trivial. The last equality follows from the fact that if K is a knot, then $\Sigma(K)$ is a rational homology sphere, see [1, Proposition 8.20].

For the proof of Item 2 refer to [10, Proposition 2.5]. □

The next theorem describes a method for obtaining genus bounds with the aid of the Casson-Gordon invariants.

Theorem 2.9 ([10, 14]). *Let $K \subset S^3$ be a knot and denote by σ_K the signature of K. Suppose that K bounds a locally flat embedded surface of genus g in the four-ball. Then, there exists a decomposition of the linking form $\lambda_K = \beta_1 \oplus \beta_2$ such that*

(1) *β_1 admits a presentation by a matrix of rank $2g$ and signature σ_K,*
(2) *β_2 is metabolic and there exists a metabolizer L of β_2 such that for any $x \in L \smallsetminus \{0\}$ we have*

$$|\sigma(K, \chi_x) + \sigma_K| \leq \eta(K, \chi_x) + 4g + 1.$$

Remark 2.10. As mentioned in the introduction, Theorem 2.9 was originally formulated in the smooth category. However, it has been understood for many years that the results of Gilmer [13] and Freedman-Quinn [12] can be used to adapt the original proof of Theorem 2.9 to work in the topological category.

2.4. *Casson–Gordon invariants of $(2, q)$-cable knots*

For a torus knot $T(2, q)$, where $q > 1$ is odd, the double branched cover $\Sigma(T(2, q))$ is the lens space $L(q, 1)$. Indeed, the torus knot $T(2, q)$ is the two-bridge knot $b(q, 1)$, see [1, Chap. 12.A], hence by [1, Proposition 12.3], $\Sigma(T(2, q)) = \Sigma(b(q, 1)) = L(q, 1)$. In particular, $H_1(\Sigma(T(2, q))) \cong \mathbb{Z}_q$ and, by [2, Proposition 1.2], the linking form $\lambda_{T(2,q)}$ is isomorphic to the linking form $(-1/q)$, see Example 2.1 for the notation.

For a general knot K, the double branched cover of its $(2, q)$-cable $K(2, q)$, can be built from $\Sigma(T(2, q))$, see [21, Lemma 4]. In particular, the following lemma, taken

from [15, Lemma 2.2], describes the most relevant properties of double-branched covers of $(2,q)$-cable knots.

Lemma 2.11. *Let K be a knot in S^3, $K(2,q)$ its $(2,q)$-cable, and $T(2,q)$ the $(2,q)$-torus knot. Choose an odd prime p and let $\xi_p = e^{2\pi i/p}$. Then:*

- *There is a map $r\colon \Sigma(K(2,q)) \to \Sigma(T(2,q)) = L(q,1)$, which is a homology isomorphism. Consequently, we can identify $H_1(\Sigma(K(2,q))) \cong \mathbb{Z}_q$.*
- *Furthermore, the map r induces an isometry of linking forms. In particular, there is a generator $l \in H_1(\Sigma(K(2,q)))$ such that $\lambda_{K(2,q)}(l,l) = -1/q$.*

The following lemma summarizes computations of Casson–Gordon invariants of $(2,q)$-cable knots for q an odd prime.

Lemma 2.12. *Let q be an odd prime.*

(1) *For any $1 \le a \le q-1$ we have*

$$\sigma(T(2,q),\chi_a) = -q + \frac{2a(q-a)}{q}, \quad \eta(T(2,q),\chi_a) = 0.$$

For $a = 0$ we have

$$\sigma(T(2,q),\chi_0) = \eta(T(2,q),\chi_0) = 0.$$

(2) *For any knot K, and any $1 \le a \le q-1$, we have*

$$\sigma(K(2,q),\chi_a) = -q + \frac{2a(q-a)}{q} + 2\sigma_K(\xi_q^a)$$

and

$$\eta(K(2,q),\chi_a) = 2\eta_K(\xi_q^a),$$

where σ_K and η_K denote the Levine–Tristram signature of K and nullity of K, respectively.

Proof. Consider first Item 1. By previous considerations, we know that $\Sigma(T(2,q))$ is the lens space $L(q,1)$, which can be presented as the $-q$ surgery on the unknot. Using the surgery formula [3, Lemma 3.1] we obtain the desired formula for $a \neq 0$. For $a = 0$, we apply Item 1 of Lemma 2.8.

The formula for the σ-invariant in Item 2 follows from [6, Corollary A.6]. The formula for the nullity follows from [6, Proposition A.10]. Although the formula in [6, Proposition A.10] is proved under the assumption that the winding number w is zero, the proof works without modification under the assumption that we work with double branched covers and $w \equiv 0 \pmod 2$. □

3. Computer-Aided Calculations of Casson–Gordon Invariants

The purpose of this section is to explain our approach towards the verification of Gilmer's criterion (Theorem 2.9) for the knot from Theorem 1.1. The main result of this section is Lemma 3.1, whose proof was done with the aid of a computer.

Let K be the knot from Theorem 1.1. Recall from Sec. 2.1, that $H_1(\Sigma(K))_q$ denotes the q-primary summand of $H_1(\Sigma(K))$, for a prime q. By Lemma 2.4, to any $x \in H_1(\Sigma(K))_q$ we associate the character

$$\chi_x \colon H_1(\Sigma(K)) \to C_q, \quad \chi_x(y) = \exp(2\pi i \lambda_K(x,y)).$$

The main result of this section is the following lemma.

Lemma 3.1. *Let K denote the knot from Theorem 1.1. Fix $q = 83, 103$. Let $x \in H_1(\Sigma(K))_q \smallsetminus \{0\}$ be an isotropic element, i.e. $\lambda_K(x,x) = 0$. There exists $0 < k < q$ such that for $y = kx$ the following inequality is satisfied*

$$|\sigma(K, \chi_y)| > 5 + \eta(K, \chi_y).$$

The strategy of the proof is to first translate Lemma 3.1 into a purely algebraic problem, which can be solved by a computer.

Problem 1. *Fix a prime number $q = 83, 103$ and let t be a positive integer relatively prime to q. For $V_q = \mathbb{Z}_q^4$ consider the following functions:*

$$s_{q,t} \colon \mathbb{Z}_q \to \mathbb{Q}, \quad \Sigma_q \colon V_q \to \mathbb{Q}, \quad Q_q \colon V_q \times V_q \to \mathbb{Z}_q$$

given by

- $s_{q,t}(x) = \begin{cases} -q + \frac{2x(q-x)}{q} + 2\sigma_{T(2,t)}(\xi_q^x), & 0 < x < q, \\ 0, & x = 0. \end{cases}$
- $\Sigma_q(x_1, x_2, x_3, x_4) = \begin{cases} s_{83,17}(x_1) - s_{83,11}(x_2) - s_{83,13}(x_3) + s_{83,1}(x_4), & q = 83, \\ s_{103,11}(x_1) - s_{103,17}(x_2) - s_{103,17}(x_3) + s_{103,13}(x_4), & q = 103. \end{cases}$
- $Q_q(x,y) = x_1y_1 - x_2y_2 - x_3y_3 + x_4y_4$,

where $\sigma_{T(2,t)}$ denotes the Levine–Tristram signature function of the torus knot $T(2,t)$ and $\xi_q = \exp(\frac{2\pi i}{q})$. Define

$$L_q = \{x \in V_q \smallsetminus \{0\} \colon Q_q(x,x) = 0\}$$

and for any $x \in L_q$ let

$$M_q(x) = \{k \colon |\Sigma_q(kx)| > 8,\ 1 \le k \le q-1\}.$$

Verify that for any $x \in L_q$, $M_q(x) \neq \emptyset$.

Lemma 3.2. *Existence of a solution to Problem 1, for $q = 83$ and $q = 103$, implies that Lemma 3.1 holds.*

Proof. Lemma 2.11 implies that

$$H_1(\Sigma(K))_q \cong \begin{cases} V_q, & q = 83, 103, \\ 0 & \text{otherwise.} \end{cases}$$

Above we choose the identification $H_1(\Sigma(K))_q \cong V_q$, for $q = 83, 103$ such that the distinguished generators (see Item 2 of Lemma 2.11) from H_1 of double branched

covers of the summands of K form a basis. Consider the primary decomposition of the linking form λ_k, as in (1),

$$(H_1(\Sigma(K)), \lambda_K) \cong \bigoplus_{q\in\{83,103\}} (H_1(\Sigma(K))_q, \lambda_{K,q}).$$

With respect to this choice of basis, the primary summands $\lambda_{K,q}$ can be identified with

$$\lambda_{K,q} \cong \begin{cases} \ell_q \oplus -\ell_q, & q = 83, 103, \\ 0 & \text{otherwise}, \end{cases}$$

where

$$\ell_q = (1/q) \oplus (-1/q),$$

refer to Example 2.1 for the notation. In particular, under this identification, if $x, y \in V_q$, then $\lambda_K(x, y) = \lambda_{K,q}(x, y) = \frac{Q_q(x,y)}{q}$, for $q = 83, 103$. Furthermore, by combining Proposition 2.8 and Lemma 2.12, we can see that for $x \in V_q$ we have

$$\sigma(K, \chi_x) = \Sigma_q(x),$$

for $x = (x_1, x_2, x_3, x_4)$. Furthermore, using inductively the formula from item 2 of Lemma 2, we can see that

$$\eta(K, \chi_x) = \#\{i\colon x_i = 0\} + \sum_{i=1}^{4} 2\eta(K_i^q, \chi_{x_i}),$$

where $K = K_1^{83}\#K_2^{83}\#K_3^{83}\#K_4^{83}\#K_1^{103}\#K_2^{103}\#K_3^{103}\#K_4^{103}$ is a decomposition as in Theorem 1.1. By applying Lemma 2.12, we see that $\eta(K_i^q, \chi_{x_i}) = 0$, for $q \in \{83, 103\}$ and $i = 1, 2, 3, 4$, therefore,

$$\eta(K, \chi_x) = \#\{i\colon x_i = 0\}.$$

In particular, since $x \neq 0$, it follows that

$$|\eta(K, \chi_x)| \leq 3.$$

Consequently, if $k \in M_q(x)$, it follows that

$$|\Sigma_q(kx)| > 8 \geq 5 + \eta(K, \chi_{kx}),$$

hence $y = kx$ satisfies the inequality in Lemma 3.1. This concludes the proof. □

Solution (Computer-aided solution to Problem 1). The solution of the problem implemented in [24] is basically a brute-force search with some optimizations.

Fix $q = 83, 103$ and recall that we denote $V_q = \mathbb{Z}_q^4$. Let $U(\mathbb{Z}_q)$ denote the multiplicative group of units of $\mathbb{Z}_q$. Observe that $U(\mathbb{Z}_q)$ acts on $V_q \setminus \{0\}$ by multiplication. Denote by W_q the quotient of $V_q \setminus \{0\}$ by the action of $U(\mathbb{Z}_q)$. Furthermore, for any

Algorithm 1. The algorithm from [24] solving Problem 1

1: **for** $q \in \{83, 103\}$ **do**
2: **for** $k = 0, 1, 2, 3$ **do**
3: **for all** $y \in V_q^k$ **do**
4: $x \leftarrow (\underbrace{0, \ldots, 0}_{3-k}, 1, y)$ ▷ Here we are generating all canonical representatives using (3).
5: **if** $Q_q(x, x) = 0$ in $\mathbb{Z}_q$ **then**
6: **for** $l \in \{1, 2, \ldots, q-1\}$ **do**
7: **if** $|\Sigma_q(lx)| > 8$ **then**
8: Print x and l and go back to line 3.
9: **end if**
10: **end for**
11: Print x and stop the program, since $M_q(x) = \emptyset$.
12: **end if**
13: **end for**
14: **end for**
15: **end for**

$x \in V_q \smallsetminus \{0\}$, we will denote by $[x]$ its image under the canonical projection to W_q. If $y \in V_q \smallsetminus \{0\}$, we say that y is in a *canonical form* if the first nonzero coordinate of y is equal to one. If $x \in L_q$ and $y \in [x]$ is in canonical form, we say that y is a *canonical representative* of $[x]$. We make the following observations:

(i) If $x \in L_q$, then the whole $U(\mathbb{Z}_q)$-orbit of x belongs to L_q.
(ii) If $x \in L_q$, then for any y in $[x]$, $M_q(x)$ and $M_q(y)$ are of the same cardinality. Indeed, write $y = kx$, for some $1 \leq k \leq q-1$, then $M_q(y) = \{k^{-1} \cdot l \colon l \in M_q(x)\}$.
(iii) For any $x \in V_q \smallsetminus \{0\}$, there exists a unique canonical representative $x_{\text{can}} \in [x]$.
(iv) Every vector x in a canonical form can be written uniquely in the form

$$x = (\underbrace{0, \ldots, 0}_{3-k}, 1, x'), \tag{3}$$

where $k = 0, 1, 2, 3$ and $x' \in \mathbb{Z}_q^k$ is arbitrary.

The above observations suggest the following simplification: it is sufficient to check the condition in Problem 1 only for vectors in canonical form.

Hence, below we present a sketch of an algorithm for solving Problem 1.

4. Genus Bounds

The purpose of this section is to prove Theorem 1.1. Let us start with a lemma giving an upper bound for the 4-genus of a certain infinite family of knots.

Let $p_1, p_2, q_1, q_2, q_3 \in \mathbb{Z}_+$ be odd primes and consider the knot

$$\begin{aligned} &K(p_1, p_2, q_1, q_2, q_3) \\ &\quad = T(2, q_1; 2, p_1) \# -T(2, q_2; 2, p_1) \# T(2, p_1) \# -T(2, q_3; 2, p_1), \\ &\qquad \# T(2, q_2; 2, p_2) \# -T(2, p_2) \# T(2, q_3; 2, p_2) \# -T(2, q_1; 2, p_2). \end{aligned}$$

Observe that taking $p_1 = 83, p_2 = 103, q_1 = 17, q_2 = 11, q_3 = 13$, we recover the knot K from Theorem 1.1.

Lemma 4.1. *For any choice of positive odd integers p_1, p_2 and non-negative odd integers q_1, q_2, q_3, we have*

$$g_4(K(p_1, p_2, q_1, q_2, q_3)) \leq 2.$$

Proof. Consider the knot

$$\begin{aligned} J(r_1, r_2, s_1, s_2) = T(2, s_1; 2, r_2) \# -T(2, s_1; 2, r_1) \# T(2, s_2; 2, r_2) \# \\ -T(2, s_2; 2, r_1), \end{aligned}$$

where r_1, r_2, s_1, s_2 are odd integers such that $r_1, r_2 > 2$ Observe, than we can write

$$K(p_1, p_2, q_1, q_2, q_3) = J_1(p_1, p_2, q_1, 0) \# J_2(p_2, p_1, q_2, q_3).$$

We will prove that $g_4(J(r_1, r_2, s_1, s_2)) \leq 1$. This would imply that

$$g_4(K(p_1, p_2, q_1, q_2, q_3)) \leq g_4(J_1(p_1, p_2, q_1, 0)) + g_4(J_2(p_2, p_1, q_2, q_3)) \leq 2,$$

as desired.

To prove that $g_4(J(r_1, r_2, s_1, s_2)) \leq 1$, we will use a modification of the argument from [15, Sec. 6]. We will construct an explicit smoothly embedded genus one surface with boundary J, for $J := J(r_1, r_2, s_1, s_2)$ in D^4. Start by performing on J band moves depicted in Fig. 1. As a result, we obtain a genus-one cobordism S_1 from J to the 2-component link L_0, which is isotopic to the $(2, 0)$-cable of the knot

$$J' = T(2, r_2) \# -T(2, r_2) \# T(2, r_1) \# -T(2, r_1). \tag{4}$$

Since J' is a slice knot (in fact a ribbon knot), let S' be the concordance from J' to the unknot. Let S_2 be the $(2, 0)$-cable of S', i.e. S_2 is the union of two-parallel copies of S' contained in some small tubular neighborhood of S'. The surface S_2, diffeomorphic to a disjoint union of two annuli, is a concordance from L_0 to the two-component unlink. By joining S_1 and S_2 along L_0 we obtain a connected, genus-one cobordism S from J to the two-component unlink. By capping off S we obtain the desired null-cobordism for J. □

We are now ready to prove Theorem 1.1.

Proof of Theorem 1.1. Observe that by [5, Theorem 1.1 and Proposition 5.4], the knot $K := K(83, 103, 17, 11, 13)$ from Theorem 1.1 is algebraically slice but not

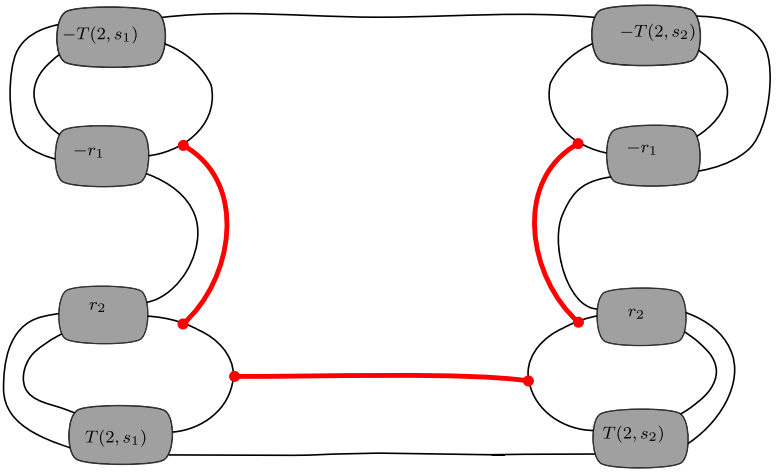

Fig. 1. Band moves (indicated in red) giving a cobordism from the knot $J(r_1, r_2, s_1, s_2)$ to the $(2,0)$-cable of the knot J', see (4).

slice. The crucial point here, is that the primes $p_1 = 83$, $p_2 = 103$, $q_1 = 11$, $q_2 = 13$, $q_3 = 17$ are pairwise distinct. By Lemma 4.1, $1 \leq g_4^{\text{top}}(K) \leq g_4(K) \leq 2$.

Suppose that $g_4^{\text{top}}(K) = 1$, we will show that this assumption contradicts Lemma 3.1. Before delving into the proof, recall from Sec. 3 that

$$H_1(\Sigma(K)) = \mathbb{Z}_{83}^4 \oplus \mathbb{Z}_{103}^4, \quad \lambda_K \cong (\ell_{83} \oplus -\ell_{83}) \oplus (\ell_{103} \oplus -\ell_{103}), \tag{5}$$

where $\ell_q = (1/q) \oplus (-1/q)$. Since K is algebraically slice, the signature of K is zero, hence by Theorem 2.9, there is a decomposition $\lambda_K = \beta_1 \oplus \beta_2$, where

- β_1 is represented by a square integral matrix A of size 2×2 and signature zero,
- β_2 is metabolic, and there is a metabolizer L for β_2, such that for any $x \in L$, we have

$$|\sigma(K, \chi_x)| \leq 5 + \eta(K, \chi_x). \tag{6}$$

We will split the argument into two cases: either $\beta_2 \neq 0$, or $\beta_2 = 0$. Let us first consider the case $\beta_2 \neq 0$. Since L is a metabolizer for β_2, by Lemma 2.3, it can be written in the form $L = L_{83} \times L_{103}$, where L_p is a metabolizer for the p-primary part $(\beta_2)_p$ of β_2, for $p = 83, 103$. Since $\beta_2 \neq 0$, we can assume without loss of generality, that $(\beta_2)_{83} \neq 0$. In particular, $L_{83} \neq 0$. Fix any $x \in L_{83} \smallsetminus \{0\}$, Lemma 3.1 shows that for some multiple $y = kx \in L_{83}$, with $1 \leq k < 83$, the condition (6) is not satisfied. Hence, we reached a contradiction, which shows that the condition $\beta_2 \neq 0$ cannot be satisfied.

If $\beta_2 = 0$, then λ_K is represented by a symmetric integral matrix A of size 2×2 and signature zero. In particular, A is a presentation matrix for $H_1(\Sigma(K))$, i.e.

$$H_1(\Sigma(K)) \cong \mathbb{Z}^2/A\mathbb{Z}^2.$$

However, the above isomorphism is not possible, since the minimal number of generators for $H_1(\Sigma(K))$ is four, see (5). This shows that neither condition $\beta_2 \neq 0$ or $\beta_2 = 0$ can hold. Consequently, our assumption $g_4^{\text{top}}(K) = 1$ lead us to a contradiction, hence $g_4^{\text{top}}(K) = g_4(K) = 2$ and the theorem is proved. □

Remark 4.2. In principle, verifying the assumptions of Theorem 2.9 requires to check all isotropic elements of $H_1(\Sigma(K))$. Since the order of $H_1(\Sigma(K))$ is $N = 83 \cdot 103 = 8549$, there are roughly $N^3 \approx 6 \cdot 10^{12}$ nonzero isotropic elements in $H_1(\Sigma(K))$. Indeed, according to [34, Chap. 3.7.2], the q-primary summand of $H_1(\Sigma(K))$ contains roughly q^3 isotropic vectors, for $q = 83, 103$. Furthermore, by modifying slightly the argument from the proof of Lemma 2.3, we can see that every isotropic element x of $H_1(\Sigma(K))$ can be written as a sum $x = x_{83} + x_{103}$, where, for $q = 83, 103$, x_q is isotropic and belongs to $H_1(\Sigma(K))_q$.

Lemma 3.1 and the proof of Theorem 1.1 show that instead of checking all isotropic vectors, it is sufficient to check only isotropic vectors belonging to primary summands of the linking form. Restricting to primary parts, reduces the number of possibilities to check roughly $83^3 + 103^3 \approx 1.6 \cdot 10^6$.

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